

SMART INDIA HACKATHON 2026



TITLE PAGE

Problem Statement ID: SIH26188 (26188)

Problem Statement Title: AI-Based Fake Identity & Document Screening System

Ministry / Organization: Ministry of Home Affairs (MHA) / Sashastra Seema Bal (SSB)

Theme: Blockchain & Cybersecurity

Category: Software

Idea Title: RakshakID: AI-Powered Multi-Layered Document Forensics & Border Screening Platform

Team ID: [Your Team ID]

Team Name: [Your Team Name]

Institute / College: [Your Institute / University Name]



Your
Team
Name

PROPOSED SOLUTION: RakshakID Platform



- ❖ • **Executive Summary:** RakshakID is an automated, high-throughput AI document forensics platform that inspects travel credentials (passports, visas, national IDs, permits) and authenticates passenger identity in under 5 seconds.
- **Multi-Document OCR & MRZ Parsing:** Simultaneously decodes ICAO Doc 9303 Machine Readable Zone (MRZ) and extracts Visual Inspection Zone (VIZ) text (Name, DOB, Expiry, Nationality, Visa Validity) with dual-stream consistency cross-checking.
- **Rule & Checksum Engine:** Computes 7-3-1 modulus-10 check-digits on MRZ strings and compares them against VIZ OCR data, issuing authority patterns, and travel validity rules to detect synthetic data manipulation.
- **Core Innovation - Multi-Layer Tampering Forensics:** Detects physical & digital forgeries via 4 specialized AI detectors: (1) Photo Replacement (boundary cut/paste & edge blending artifacts), (2) Text Manipulation (font inconsistency & character alignment warping), (3) Stamp Forgery (circular contour distortion & ink hue analysis), and (4) Error Level Analysis (ELA) to expose digital recompression.
- **1:1 Live Biometric Verification:** Extracts facial embeddings from the document photo using DeepFace (ArcFace backbone) and matches them against live traveler webcam capture with real-time passive liveness/blink detection.
- **Unified Risk Scoring Dashboard:** Generates an explainable 0–100 Risk Score (Green=Cleared, Amber=Review, Red=Detain) with visual forensic bounding boxes and exportable digital investigation dossiers.

Your
Team
Name

TECHNICAL APPROACH & PIPELINE ARCHITECTURE



- **Full-Stack Architecture:** Frontend: Next.js / React with Tailwind CSS, HTML5 Canvas for forensic heatmaps, WebRTC for webcam capture. Backend: Python FastAPI async server for low-latency multi-threaded image processing.
- **Step 1 - Image Preprocessing:** OpenCV deskewing, perspective rectification (homography), adaptive lighting compensation, and unsharp masking to enhance faded text and stamps on physical IDs.
- **Step 2 - Dual OCR & MRZ Extraction:** Dedicated passporteye / pytesseract pipeline to extract MRZ TD1/TD2/TD3 strings, combined with Google Vision / Gemini Vision API for high-accuracy VIZ text parsing.
- **Step 3 - ICAO 9303 Verification:** Cryptographic 7-3-1 weight modulus-10 checksum validation on document number, DOB, and expiration date; cross-references MRZ against VIZ text.
- **Step 4 - AI Tampering & Forgery Engine:** Error Level Analysis (ELA) for detecting JPEG re-compression discrepancies; ResNet/ViT classifier for photo border splicing; EXIF forensic parser for editing software signatures (Photoshop/GIMP).
- **Step 5 - 1:1 Biometric Match & Liveness:** DeepFace (ArcFace model) compares document face photo to live traveler camera (cosine similarity threshold = 0.68). MediaPipe landmark tracking verifies eye blink & head pose to thwart photo/screen spoofing.
- **Step 6 - Decision & Audit Trail:** Aggregates module confidence scores into a single Risk Index; logs timestamped events into PostgreSQL with encrypted hash trails and generates 1-click forensic PDF dossiers.

Your
Team
Name

FEASIBILITY, VIABILITY & RISK MITIGATION



- **Technical Feasibility:** Employs open, battle-tested computer vision & AI frameworks (OpenCV, ArcFace, Tesseract, ELA). Inference latency is <1.5 seconds on standard x86 CPU workstations without requiring costly GPU servers.
- **Operational Feasibility:** Integrates smoothly into current SSB immigration booths with existing document flatbed scanners and USB webcams. Zero specialized training needed; clear intuitive color-coded risk dashboard.
- **Economic Viability:** 100% software-based architecture with minimal upfront capital expenditure; avoids expensive proprietary spectral/IR hardware scanners, ensuring cost-effective deployment across hundreds of border posts.
- **Challenge 1: Poor Scan Quality & Glare at Border:** Mitigation: Built-in automated image enhancement pipeline utilizing Contrast Limited Adaptive Histogram Equalization (CLAHE) and specular reflection reduction algorithms before OCR processing.
- **Challenge 2: Intermittent Connectivity at Remote Outposts:** Mitigation: Hybrid Edge-First Architecture: OCR, MRZ checksums, and 1:1 facial matching execute fully offline on local PCs. Scans and logs automatically sync to central MHA databases once connectivity restores.
- **Challenge 3: Sophisticated Biometric Spoofing (Print/Screen):** Mitigation: Multi-frame temporal liveness analysis (micro-blinking, facial depth heuristics) blocks printed photographs and mobile screen replays.

Your
Team
Name

IMPACT, BENEFITS & NATIONAL SECURITY



- **Target Stakeholders:** Sashastra Seema Bal (SSB), Bureau of Immigration (BoI), Ministry of Home Affairs (MHA), State Police Checkpoints, and honest cross-border travelers.
- **90% Faster Passenger Clearance:** Slashes manual verification latency from 3–5 minutes down to <5 seconds per traveler, eliminating congestion and long queues at high-density transit terminals.
- **Superhuman Tamper & Forgery Detection:** Detects micro-level font splicing, altered birthdates, swapped photos, and fake seals that easily deceive human visual inspection during peak fatigue hours.
- **Standardized & Objective Screening:** Replaces subjective officer judgment with objective, data-driven risk scoring, establishing consistent border security protocols across all land border outposts.
- **Tamper-Proof Legal Evidence Dossier:** Automatically compiles timestamped forensic reports with highlighted anomaly bounding boxes, creating legally admissible evidence chains for prosecutions and intelligence sharing.
- **National Security Fortification:** Prevents unauthorized cross-border infiltration, human trafficking, and fake passport syndicates across India's open borders (Indo-Nepal & Indo-Bhutan), directly reinforcing national sovereignty.

Your
Team
Name

RESEARCH, STANDARDS & REFERENCES



- **International Standards (ICAO Doc 9303):** Specifications for Machine Readable Travel Documents (MRTDs), Parts 3, 4 & 7: Standardized MRZ formats (TD1, TD2, TD3) and 7-3-1 weight modulus-10 check digit algorithms.
- **Biometric Verification Standards (ISO/IEC 19794-5):** Information Technology — Biometric data interchange formats — Part 5: Face image data quality, resolution, and landmark standards.
- **Digital Image Forensic Research (ELA):** Krawetz, Neal (2007). 'A Picture's Worth... Digital Image Analysis and Error Level Analysis (ELA)', Hacker Factor Solutions — Foundations of JPEG compression artifact forensics.
- **Deep Facial Embeddings (ArcFace):** Deng, J., Guo, J., Xue, N., & Zafeiriou, S. (2019). 'ArcFace: Additive Angular Margin Loss for Deep Face Recognition', IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR).
- **Document Tampering & Splicing Localization:** Bappy, J. H., Roy-Chowdhury, A. K., et al. (2019). 'Exploiting Spatial Structure for Localizing Manipulated Image Regions', IEEE Transactions on Information Forensics and Security.
- **National Guidelines & Policy Frameworks:** Bureau of Police Research and Development (BPR&D) Document Examination Manual; Ministry of Home Affairs (MHA) Integrated Border Management System (IBMS) Guidelines.